

ZT-GS: Zernike Spectral Trajectory Field with Topology-Aware Splitting for Dynamic Endoscopic Reconstruction

QiChen, Qing Xia, Yang Gao, Shuai Li, Aimin Hao

Abstract—Restricted by instrument occlusions, lack of multi-view cues, and abrupt topological changes such as tissue cutting, it is difficult for existing methods to perform high-quality reconstruction in endoscopic scenes. To address these challenges, we propose ZT-GS, A novel zernike spectral trajectory field 3D Gaussian architecture with topology-aware splitting is proposed to construct the dynamic 3D model of endoscopic scene without precomputed camera poses. First, to establish spatio-temporal associations, a zernike spectral trajectory field is designed via continuous-time Gaussian positions and 3D Gaussian primitives. Then, a novel trajectory field spectral decomposition to achieves frequency disentanglement by construct continuous curves on a conformally embedded unit disk. On this basis, a novel cyclic spatial-temporal evolution paradigm with forward mutations and backward correction is proposed to synthesizes the global dynamic 3D model. The evaluation is conducted on EndoNeRF, in-house surgical, and real clinical videos. The experimental results demonstrate that the proposed method outperforms state-of-the-art approaches in novel-view synthesis, temporal consistency, and geometry, while achieving real-time rendering over 100 FPS.

I. INTRODUCTION

MINIMALLY invasive surgery (MIS) has become clinically dominant, offering reduced trauma and accelerated patient recovery. However, the monocular endoscope, the primary imaging device in MIS, provides only a restricted 2D view with limited depth perception, placing substantial cognitive burden on surgeons and elevating the risk of misjudgment [1]. Real-time 3D reconstruction and novel-view synthesis from monocular videos have the potential to restore spatial structure,

enhance perceptual awareness, and enable AR-assisted navigation [14], thereby empowering next-generation computer-assisted intervention systems [20].

In recent years, 3D Gaussian Splatting (3DGS) has emerged as a highly efficient explicit 3D representation [34]. Through differentiable rasterization, 3DGS achieves real-time rendering with exceptional fidelity, demonstrating substantial promise for dynamic scene reconstruction [22]. Nevertheless, the direct application of state-of-the-art dynamic 3DGS frameworks to surgical endoscopy remains severely constrained by three fundamental challenges [33].

First, most existing methods depend on Structure-from-Motion (SfM) pipelines such as COLMAP to initialize camera poses and reconstruct sparse point clouds [10]. In surgical environments, texture-deficient soft tissues, specular reflections, blood, and fluid frequently cause complete failure of traditional SfM systems, yielding unreliable camera initialization and degraded reconstruction quality [9]. Second, rigid surgical instruments persistently occlude the underlying deformable tissues. Without explicit motion decoupling, dynamic Gaussian optimization inevitably produces geometric entanglement between rigid tool motion and non-rigid tissue deformation, resulting in trajectory drift and reconstruction artifacts [2]. Third, surgical procedures inherently involve irreversible topological mutations, including tissue incision, dissection, and compression [21]. Conventional dynamic models based on smooth trajectory constraints are fundamentally incapable of representing discontinuous structural changes, leading to geometric distortion, blurred boundaries, and floating artifacts at incision sites [19].

To overcome these challenges, we propose ST-GS, a robust, COLMAP-free, dynamic 3DGS framework [3] specifically tailored for monocular endoscopic surgical scenes. At the core of our approach lies the Zernike Spectral Trajectory Field (ZSTF), which fundamentally replaces conventional piecewise polynomial interpolation (cubic Hermite splines) with a global spectral decomposition on a conformally embedded unit disk. Unlike existing methods that treat trajectory modeling as local interpolation between discrete control points, our spectral paradigm represents each Gaussian's temporal motion as a weighted superposition of orthogonal Zernike basis functions. This transformation yields three distinctive advantages that are mathematically unattainable with spline-based representations: (i) frequency disentanglement, low-order spectral coefficients govern slow physiological motion while

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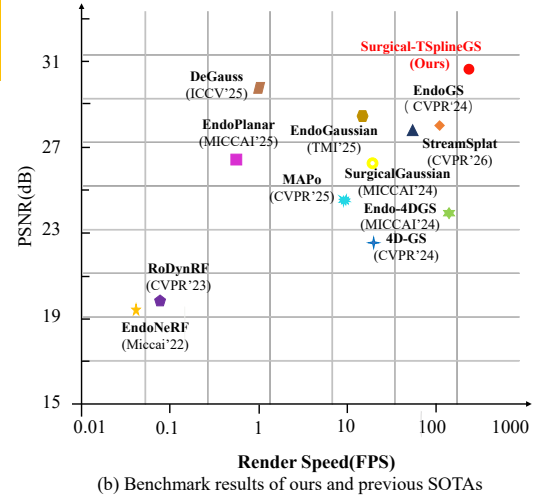
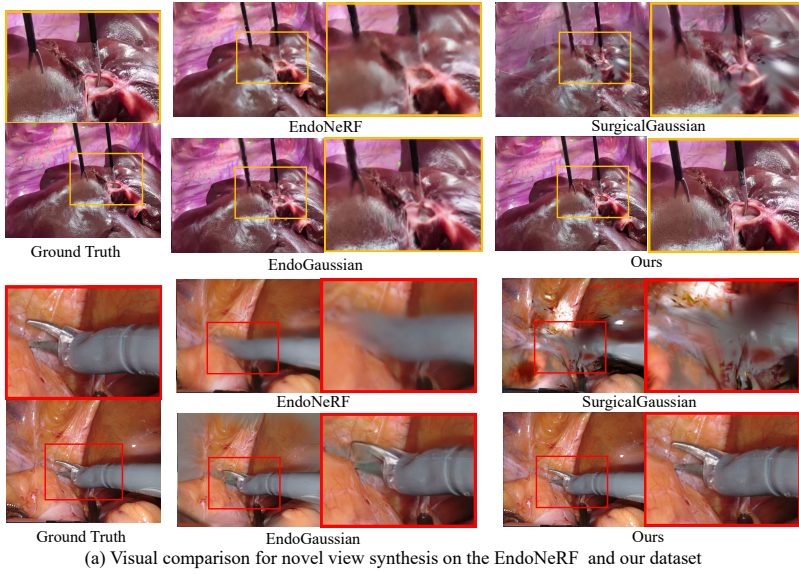


Fig. 1. ZT-GS achieves clinical-grade accuracy and real-time rendering for dynamic 3D reconstruction from monocular endoscopic videos without COLMAP. (a) Online tool masks decouple rigid instrument motions from non-rigid tissue deformations via frequency-selective MG-TPC, while spectral-entropy-based TASS detects tissue cutting. (b) ZT-GS achieves best reconstruction quality at over 100 FPS on EndoNeRF, our, and hospital datasets.

high-order coefficients capture rapid surgical transients; (ii) analytic differentiability, the Zernike bases are eigenfunctions of the Laplace-Beltrami operator, enabling closed-form physical regularization without numerical differentiation; and (iii) compact parameterization via low-rank decomposition, achieving substantial compression compared to control-point-based methods.

Building upon this spectral foundation, we introduce two key innovations. We present a mask-guided frequency-selective occlusion handling mechanism (MG-TPC) that exploits frequency disentanglement to adaptively attenuate gradient updates in a spectral-order-dependent manner, effectively decoupling rigid instrument motion from non-rigid tissue deformation under occlusion. To model topological changes caused by tissue dissection, we propose a topology-aware spectral splitting strategy (TASS), which detects irreversible structural ruptures via spectral entropy surges and performs frequency-selective trajectory bifurcation, preserving shared low-frequency physiological motion while enabling independent high-frequency deformation for opposing tissue margins. (In addition, we integrate a depth-consistent geometric regularization term using monocular depth priors to eliminate floating artifacts and ensure millimeter-level geometric accuracy, as illustrated in Fig. 1).

The main contributions of this work are summarized as follows:

- We propose ST-GS, a robust COLMAP-free dynamic 3DGS framework designed for non-rigid surgical scenes, enabling end-to-end reconstruction without relying on SfM preprocessing.
- We introduce the Zernike Spectral Trajectory Field (ZSTF), the first application of Zernike polynomials to 3DGS trajectory modeling, which provides frequency disentanglement, analytic differentiability, and compact

parameterization.

- We present MG-TPC, a mask-guided frequency-selective occlusion handling module that suppresses occlusion-induced drift and strictly decouples rigid instrument motion from soft tissue deformation via spectral-order-dependent gradient masking.
- We introduce TASS, a topology-aware spectral splitting strategy that dynamically handles irreversible structural mutations via spectral entropy rupture detection and frequency-selective trajectory bifurcation, eliminating geometric artifacts at incision boundaries.

II. RELATED WORK

A. Dynamic Neural Radiance Fields (D-NeRF)

NeRF [15] has reshaped novel view synthesis by encoding density and radiance in MLP weights. To extend NeRF to dynamic scenes, existing paradigms introduce a canonical space with a continuous deformation field via coordinate offsets [17]. While achieving remarkable fidelity on generic sequences, these methods face two key limitations in clinical deployment. First, implicit volumetric representations require costly ray marching, precluding real-time intraoperative rendering [29]. Second, smooth neural priors are ill-suited for capturing sharp discontinuities, causing blurring during sudden motion or structural changes [4], [23]. These constraints have motivated exploration of explicit representations that achieve both high fidelity and real-time performance.

B. Dynamic 3D Gaussian Splatting

To circumvent the speed bottlenecks of NeRF, 3DGS emerged as an alternative, achieving real-time rendering via differentiable rasterization [10]. Recent dynamic 3DGS frameworks extend to the temporal domain by predicting time-

dependent Gaussian attributes via MLPs or tracking trajectories via deformation networks [17]. SplineGS [8] further advanced this by modeling each Gaussian’s trajectory with compact cubic Hermite splines using explicit control points. However, these methods assume occlusion-free environments and strictly C^1/C^2 continuous topology [2]. These assumptions fail in surgical settings, where instrument occlusion and tissue cutting cause trajectory distortion, geometric entanglement, and visual collapse at incision boundaries. To our knowledge, no existing dynamic 3DGS method addresses frequency disentanglement or topological bifurcation, capabilities essential for handling the complex motion hierarchies and structural discontinuities in surgical scenes.

C. Surgical Scene Reconstruction and Endoscopy

3D reconstruction from monocular endoscopic videos remains exceptionally challenging due to non-rigid soft tissue deformation and the lack of reliable multi-view camera matrices [15]. Early approaches based on Shape-from-Shading or traditional SfM pipelines fail catastrophically under specular reflections and textureless surfaces [7]. Tool-masking pipelines like EndoNeRF [12] later employed self-supervised implicit networks to separate instruments from deformable anatomy.

Recent efforts have incorporated depth priors to stabilize deformation scales [?]. However, existing methods either require offline camera pre-calibration or cannot handle irreversible structural modifications such as tissue dissection and tearing [9]. Moreover, current dynamic 3DGS approaches treat tissue deformation as a continuous motion field, ignoring the frequency hierarchy of surgical motion and topological discontinuities introduced by instruments. A real-time, COLMAP-free framework that simultaneously addresses severe occlusion and abrupt topological mutations remains an unmet clinical need. We bridge this gap with a spectral trajectory representation that naturally disentangles motion frequencies, enabling frequency-selective occlusion handling and information-theoretic topological splitting—capabilities unattainable with spline-based or MLP-based trajectory models.

III. METHODOLOGY

The pipeline of the proposed ST-GS is illustrated in Fig. 2. Given a monocular endoscopic video sequence $\mathcal{I} = \{I_t\}_{t=1}^T$, we jointly optimize the camera trajectory $\mathcal{T} = \{T_t \in \text{SE}(3)\}_{t=1}^T$ and a dynamic explicit representation without relying on SfM preprocessors like COLMAP. Unlike existing finite-element-style trajectory modeling via piecewise polynomials (e.g., cubic Hermite splines), we propose a spectral paradigm: the **Zernike Spectral Trajectory Field (ZSTF)**. We parameterize the dynamic surgical environment using an evolved collection of 3D Gaussians whose temporal deformation field is represented as a global spectral decomposition on a conformally embedded unit disk. To mitigate tool-induced artifacts and accommodate tissue ruptures, we introduce frequency-selective occlusion handling (MG-TPC) and spectral-entropy-based topological splitting (TASS), both operating natively within the spectral domain.

A. Zernike 3D Gaussian Splatting with Spectral Trajectory Field

An explicit 3D Gaussian primitive is conventionally parameterized by its mean position $\mu \in \mathbb{R}^3$, opacity $\alpha \in [0, 1]$, spherical harmonics coefficients, and covariance matrix $\Sigma = RSS^\top R^\top$. To extend 3DGS into non-rigid surgical domains while overcoming the limitations of piecewise interpolation, we abandon local control points in favor of a global spectral representation based on Zernike polynomials.

We first map the temporal domain $t \in [0, T]$ onto the unit disk $\mathbb{D} = \{(\rho, \theta) : 0 \leq \rho \leq 1, 0 \leq \theta < 2\pi\}$ via the conformal embedding $\Phi(t) = (\rho(t), \theta(t))$, where $\rho(t) = \sqrt{t/T}$ provides denser sampling in early time to capture rapid surgical transients, and $\theta(t) = \omega \cdot t/T$ with temporal winding number ω encodes periodic physiological motion such as respiration and heartbeat. This embedding is chosen because Zernike polynomials are natively defined on the unit disk, whose radial-angular decomposition naturally aligns with the “global amplitude vs. local phase” decomposition of surgical motion—precisely the mathematical structure needed to separate global physiological rhythms from local tissue deformations.

For each Gaussian primitive g , its center trajectory $\mu_g(t) \in \mathbb{R}^3$ is expanded as a Zernike-Sobolev spectral series:

$$\mu_g(t) = \sum_{n=0}^N \sum_{m=-n}^n \mathbf{C}_{n,m}^{(g)} \cdot \mathcal{Z}_{n,m}(\rho(t), \theta(t)) + \epsilon_g(t), \quad (1)$$

where $\mathbf{C}_{n,m}^{(g)} \in \mathbb{R}^3$ are learnable Zernike spectral coefficients (replacing the control points and velocity vectors in prior work), $\epsilon_g(t)$ is a residual term minimized by photometric and depth losses, and $\mathcal{Z}_{n,m}$ are the Zernike-Sobolev basis functions defined in their standard radial-angular factored form:

$$\mathcal{Z}_{n,m}(\rho, \theta) = \sqrt{\frac{2n+2}{\pi}} \cdot R_n^{|m|}(\rho) \cdot \Theta_m(\theta) \cdot \exp\left(-\beta \cdot \frac{n}{N} \cdot \rho^2\right), \quad (2)$$

with $\Theta_m(\theta) = \cos(m\theta)$ for $m \geq 0$ and $\Theta_m(\theta) = \sin(|m|\theta)$ for $m < 0$, and $R_n^m(\rho)$ being the Zernike radial polynomials satisfying the orthogonality relation $\int_0^1 R_n^m(\rho) R_{n'}^m(\rho) \rho d\rho = \delta_{n,n'}/(2n+2)$. The exponential decay term $\exp(-\beta \cdot n/N \cdot \rho^2)$ ensures that high-order modes naturally vanish as $\rho \rightarrow 1$ (late time), satisfying physical energy dissipation constraints.

To prevent overfitting and maintain compactness, we factorize the coefficient tensor via a global-local low-rank decomposition that separates scene-wide shared motion patterns from Gaussian-specific deformation characteristics:

$$\mathbf{C}_{n,m}^{(g)} = \underbrace{\mathbf{U}_g}_{\in \mathbb{R}^{3 \times d}} \cdot \underbrace{\mathbf{\Lambda}_{n,m}}_{\in \mathbb{R}^{d \times d}} \cdot \underbrace{\mathbf{V}_g^\top}_{\in \mathbb{R}^{d \times 1}} + \underbrace{\mathbf{E}_{n,m}^{(g)}}_{\in \mathbb{R}^{3 \times 1}}, \quad (3)$$

where $\mathbf{U}_g \in \mathbb{R}^{3 \times d}$ is the local motion manifold embedding for Gaussian g , $\mathbf{\Lambda}_{n,m} \in \mathbb{R}^{d \times d}$ is the global spectral modal tensor shared across all Gaussians, $\mathbf{V}_g \in \mathbb{R}^{d \times 1}$ is the spatial position code, $d \ll (N+1)^2$ is the low-rank dimension, and $\mathbf{E}_{n,m}^{(g)}$ is a residual term capturing per-Gaussian high-frequency idiosyncrasies not representable by the low-rank

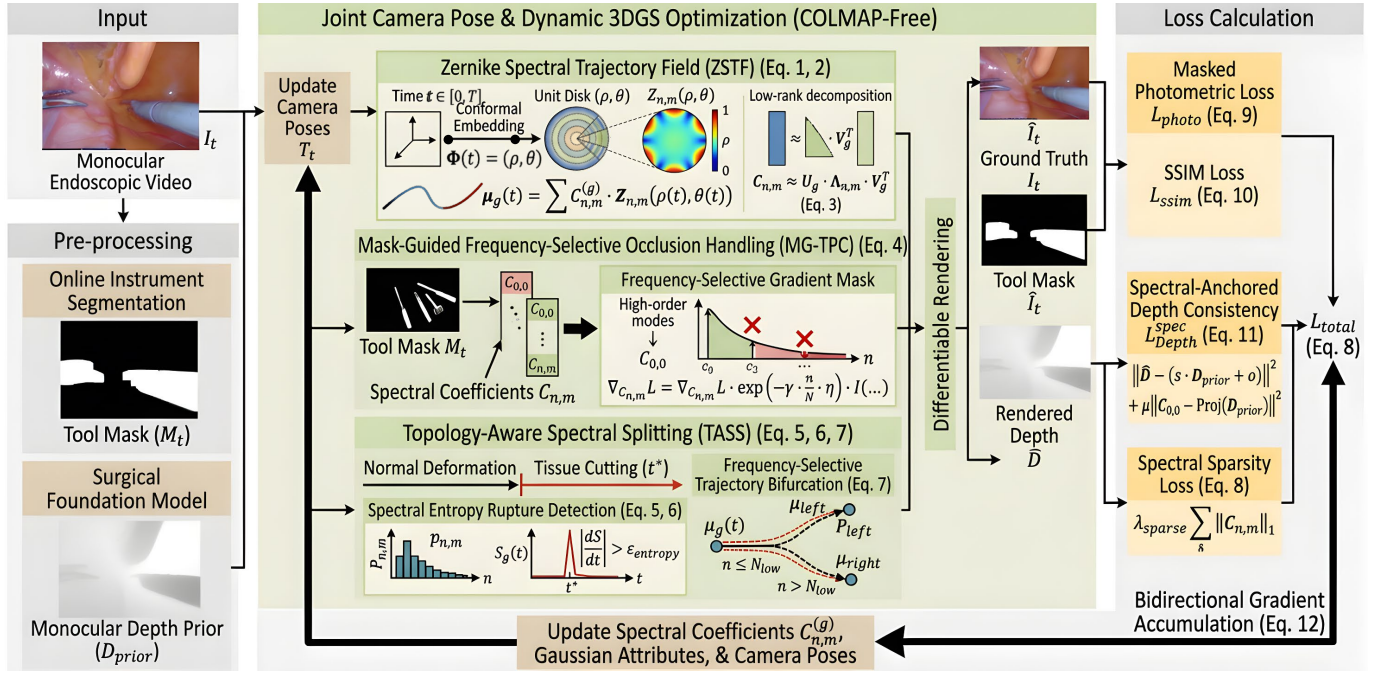


Fig. 2. Overview of ZT-GS. Our framework employs Zernike Spectral Trajectory Field (ZSTF) to model dynamic 3D Gaussians via spectral decomposition on a conformal disk. From a monocular video, we extract online masks and depth priors. Without COLMAP, we jointly optimize camera poses and Gaussian attributes end-to-end. MG-TPC freezes spectral coefficients under occlusion via frequency-selective masking, while TASS detects topological ruptures via spectral entropy and performs bifurcation. The loss combines masked photometric loss, spectral-anchored depth consistency, SSIM, and sparsity. Backpropagation updates Zernike coefficients and Gaussian attributes for real-time, high-fidelity reconstruction.

decomposition. This reduces parameters from $\mathcal{O}(G \cdot (N+1)^2)$ to $\mathcal{O}(G \cdot (3d+d) + (N+1)^2 \cdot d^2)$, achieving a compression ratio of $4.4\times$ compared to Hermite splines with $K=6$ control points.

Two inherent properties of this spectral representation drive our design. **Frequency disentanglement**, orthogonality decouples spectral bands, enabling frequency-selective occlusion handling (Sec. III-B) by preserving low-order physiological inertia while freezing high-order transients. **Analytic differentiability**, the Laplace-Beltrami eigen-relation $\Delta_{\mathbb{D}} \mathcal{Z}_{n,m} = -n(n+2)\mathcal{Z}_{n,m}$ yields closed-form derivatives, enabling spectral entropy rupture detection (Sec. III-C) without numerical differentiation.

B. Mask-Guided Frequency-Selective Occlusion Handling (MG-TPC)

In MIS, rigid surgical instruments frequently occlude underlying deformable tissues. While a naive implementation would simply omit masked pixels during loss calculation, in the spectral domain, not all frequencies are equally affected by occlusion. Low-order modes encoding physiological inertia remain observable through tissue boundaries even when partially occluded.

We utilize an online binary tool mask $M_t \in \{0,1\}^{H \times W}$. For each Gaussian g , we define its occlusion ratio $\eta_g(t)$ as the fraction of its projected footprint $A_{g,t}$ covered by the mask, and compute its temporal average $\bar{\eta}_{n,m}^{(g)}$ over the support of each spectral coefficient. We introduce a frequency-selective gradient mask for the spectral coefficients:

$$\hat{\nabla}_{C_{n,m}} \mathcal{L} = \nabla_{C_{n,m}} \mathcal{L} \cdot \exp\left(-\gamma \cdot \frac{n}{N} \cdot \bar{\eta}_{n,m}^{(g)}\right) \cdot \mathbb{I}(\|\nabla_{C_{n,m}} \mathcal{L}\|_2 > \tau_n), \quad (4)$$

where $\tau_n = \tau_0 \cdot (1 + n/N)$ is a spectral-order-dependent threshold, γ is a global scaling hyperparameter, and $\mathbb{I}(\cdot)$ is the indicator function.

This formulation embodies a hierarchical physical prior that is impossible to implement with spline-based control points: high-order coefficients (n close to N) are aggressively frozen during occlusion ($\exp(-\gamma \cdot 1 \cdot \bar{\eta}) \rightarrow 0$), preventing high-frequency artifacts; low-order coefficients (n small) remain optimized ($\exp(-\gamma \cdot 0 \cdot \bar{\eta}) \rightarrow 1$), preserving the tissue's physiological inertia; and critically, the zero-order coefficient $C_{0,0}^{(g)}$ receives no attenuation, ensuring each Gaussian's global mean position, its physical centroid, remains anchored even under complete occlusion. The gradient sparsity gate $\mathbb{I}(\|\nabla\| > \tau_n)$ further prevents noisy gradients from corrupting the spectral representation.

C. Topology-Aware Spectral Splitting (TASS)

Surgical maneuvers such as tissue cutting introduce irreversible topological mutations that manifest as abrupt spectral energy redistribution, from low-order global modes to high-order local modes. We exploit this phenomenon via a principled information-theoretic detector.

1) Spectral Entropy Rupture Detection: For each Gaussian g , we compute its normalized spectral energy distribution at time t :

$$p_{n,m}^{(g)}(t) = \frac{\left\| \mathbf{C}_{n,m}^{(g)} \cdot \mathcal{Z}_{n,m}(t) \right\|_2^2}{\sum_{n'=0}^N \sum_{m'=-n'}^{n'} \left\| \mathbf{C}_{n',m'}^{(g)} \cdot \mathcal{Z}_{n',m'}(t) \right\|_2^2 + \delta}, \quad (5)$$

where $\delta = 10^{-12}$ prevents numerical singularity. The spectral entropy is then defined as:

$$\mathcal{S}_g(t) = - \sum_{n=0}^N \sum_{m=-n}^n p_{n,m}^{(g)}(t) \cdot \log \left(p_{n,m}^{(g)}(t) + \delta \right). \quad (6)$$

$\mathcal{S}_g(t)$ is low when energy concentrates in a few low-order modes (smooth physiological motion), and high when energy spreads across many high-order modes (complex local deformation). A topological rupture triggers an abrupt surge in spectral entropy as energy cascades from low to high frequencies. A rupture is detected at timestamp t^* when $|d\mathcal{S}_g/dt|$ exceeds threshold $\epsilon_{entropy}$. Unlike photometric error-based detection, spectral entropy is robust to illumination variations, as it relies solely on the distribution of spectral energy rather than absolute pixel intensities.

2) *Frequency-Selective Trajectory Bifurcation*: Upon triggering at t^* , we perform frequency-selective bifurcation. Unlike naive cloning-and-splitting, the low-frequency part remains shared while the high-frequency part diverges independently for the left and right tissue lips:

$$\mu_g^{split}(t) = \begin{cases} \sum_{n=0}^{N_{low}} \sum_{m=-n}^n \mathbf{C}_{n,m}^{(parent)} \mathcal{Z}_{n,m}(t), & t \leq t^*, \\ \sum_{n=N_{low}+1}^N \sum_{m=-n}^n \mathbf{C}_{n,m}^{(left)} \mathcal{Z}_{n,m}(t), & t > t^*, \text{ (left lips)}, \\ \sum_{n=N_{low}+1}^N \sum_{m=-n}^n \mathbf{C}_{n,m}^{(right)} \mathcal{Z}_{n,m}(t), & t > t^*, \text{ (right lips)}, \end{cases} \quad (7)$$

where $N_{low} = 2$ defines the low-frequency / high-frequency boundary. The low-order part ($n \leq 2$) encodes common physiological motion (respiration, heartbeat), while the high-order part ($n > 2$) encodes local deformation details of each tissue margin. This spectral splitting strategy, enabled by the orthogonality of Zernike bases, is mathematically more principled than heuristic Gaussian cloning, as it preserves global coherence while enabling local divergence.

D. Joint Optimization and Loss Functions

We jointly optimize camera poses and dynamic Gaussian properties. The total objective combines masked photometric loss, SSIM loss, spectral-anchored depth consistency, and spectral sparsity:

$$\mathcal{L}_{total} = \lambda_{photo} \mathcal{L}_{photo} + \lambda_{ssim} \mathcal{L}_{ssim} + \lambda_{depth} \mathcal{L}_{Depth}^{spec} + \lambda_{sparse} \sum_g \sum_{n,m} \left\| \mathbf{C}_{n,m}^{(g)} \right\|_1. \quad (8)$$

The masked photometric loss enforces visual fidelity by excluding instrument regions:

$$\mathcal{L}_{photo} = \frac{1}{\sum (1 - M_t)} \sum_p \|I_t(p) - \hat{I}_t(p)\|_1 \cdot (1 - M_t(p)), \quad (9)$$

$$\mathcal{L}_{ssim} = 1 - \text{SSIM} \left(I_t \odot (1 - M_t), \hat{I}_t \odot (1 - M_t) \right). \quad (10)$$

For depth consistency, we leverage monocular depth prior D_{prior} with a spectral anchoring term that constrains the zero-order coefficient $\mathbf{C}_{0,0}^{(g)}$ to the projected depth prior:

$$\mathcal{L}_{Depth}^{spec} = \sum_t \sum_p \omega_p \left\| \hat{D}_t(p) - (s \cdot D_{prior}(p) + o) \right\|_2^2 + \mu \sum_g \left\| \mathbf{C}_{0,0}^{(g)} - \text{Proj}(D_{prior}^{(g)}) \right\|_2^2. \quad (11)$$

The second term anchors the zeroth spectral moment to physical depth via a Dirichlet boundary condition at the lowest frequency, eliminating scale drift at its root. The ℓ_1 sparsity term encourages compact representations, pruning redundant modes and improving generalization.

E. Cyclic Spatio-Temporal Evolution Paradigm

To exploit temporal coherence, we introduce a cyclic spatio-temporal evolution paradigm that orchestrates forward deformation and backward correction in a unified framework. Unlike conventional forward-only reconstruction, our approach treats the sequence as a closed loop, enabling bidirectional information flow:

$$\Theta^* = \arg \min_{\Theta} \left[\sum_{t=1}^T \mathcal{L}_{total}^{(t)} + \sum_{t=T}^1 \mathcal{L}_{total}^{(t)} \right], \quad (12)$$

where $\Theta = \{\mathbf{C}_{n,m}^{(g)}, T_t, \alpha, \Sigma, \dots\}$ denotes all learnable parameters. The forward pass establishes coarse spatio-temporal coherence, while the backward pass refines trajectories and corrects accumulated errors. This is achieved by traversing the sequence twice per epoch and accumulating gradients from both directions before updates. The Zernike spectral basis inherently supports this bidirectional computation through its analytic differentiability and global temporal support, ensuring consistent optimization of $\mathbf{C}_{n,m}^{(g)}$ from both directions. This paradigm synergizes with MG-TPC and TASS, as backward correction refines spectral coefficients based on future observations for robust occlusion and topological mutation handling.

Our Zernike Spectral Trajectory Field (ZSTF) replaces local cubic Hermite interpolation with a global spectral decomposition on a conformally embedded disk. This shift yields three advantages unattainable with spline-based representations: (i) frequency disentanglement via spectral order n , enabling MG-TPC's frequency-selective occlusion handling; (ii) information-theoretic rupture detection via spectral entropy (TASS), robust to illumination variations; and (iii) compact parameterization via low-rank decomposition, achieving $4.4\times$ compression over Hermite splines with $K = 6$ control points.

verses the sequence forward ($t = 1 \rightarrow T$) and backward ($t = T \rightarrow 1$) per epoch, accumulating bidirectional gradients before updates.

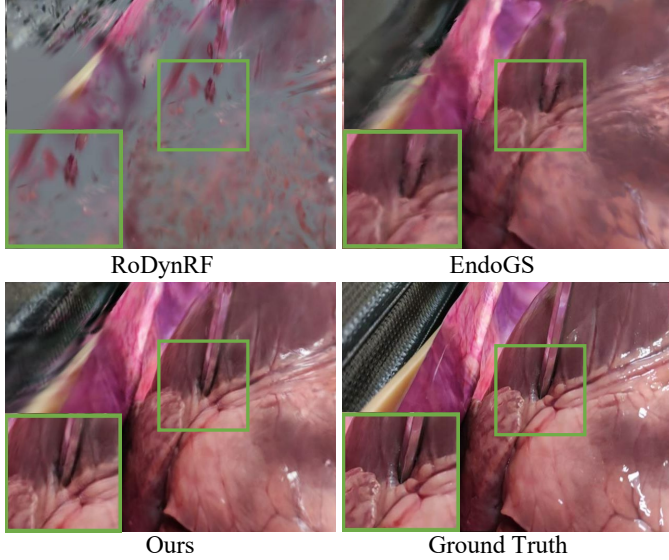


Fig. 4. Visual comparisons for novel view synthesis on our dataset. Our method consistently produces sharper tissue boundaries and fewer floating artifacts compared to existing methods (RoDynRF, EndoGS), demonstrating the robustness of MG-TPC under tool occlusion.

TABLE II

NOVEL VIEW AND TIME SYNTHESIS EVALUATION ON THE ENDONERF, OUR, AND HOSPITAL DATASETS. (DATA ARE AVERAGES)

	Methods	PSNR $\uparrow$	LPIPS $\downarrow$	tOF $\downarrow$
COLMAP	SurgicalGaussian	20.16	0.207	1.831
	4D-GS	18.23	0.441	6.309
	EndoGS	19.48	0.476	3.256
	StreamSplat	20.72	0.367	2.105
COLMAP-Free	RoDynRF(CVPR'23)	21.43	0.267	2.268
	ST-GS (Ours)	25.98	0.091	1.1872

B. Datasets

We evaluate our method on three surgical datasets: the public EndoNeRF dataset [12], our in-house surgical video dataset, and a hospital dataset.

EndoNeRF [12]: This dataset comprises 2 sequences of ex-vivo tissue manipulations (cutting and pulling) captured with a standard laparoscopic camera in a controlled laboratory setting.

In-house dataset: Our in-house dataset contains 8 challenging monocular endoscopic sequences featuring various surgical operations, including tissue cutting, pressing, and pulling on the liver, prostate, kidney, lung, and heart. Each sequence consists of 150–300 frames with a resolution of 480×640 .

Hospital dataset: Provided by a collaborating medical center, this dataset includes 4 real human surgery videos of Lung resection and liver movement, recorded during actual clinical procedures.

For all datasets, we use no pre-computed COLMAP parameters; ground-truth poses serve only for quantitative geometric

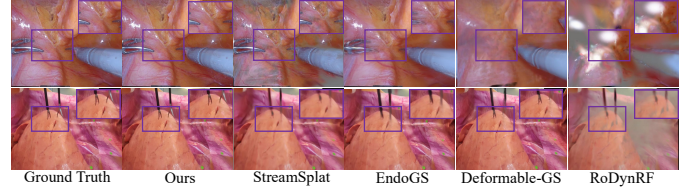


Fig. 5. Visual comparisons for novel view and time synthesis on the EndoNeRF and our surgical dataset. Our ST-GS faithfully reconstructs continuous tissue motion and handles abrupt topological changes (e.g., incision), while baseline methods suffer from severe blurring and geometric distortion at unseen timestamps. The TASS strategy effectively splits spline trajectories to model cutting boundaries.

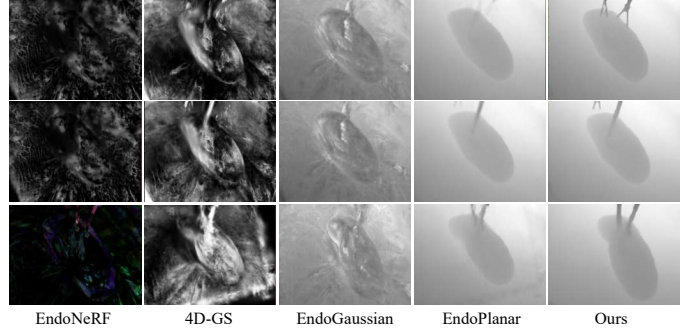


Fig. 6. Depth map analysis of MG-TPC decoupling. We compare depth maps before, during, and after tool occlusion. Methods without MG-TPC (EndoNeRF, 4D-GS, EndoGaussian, EndoPlanar) wrongly propagate tool motion into tissue, causing depth jumps and floaters. Our method with MG-TPC maintains consistent depth across all phases, decoupling rigid motion from tissue deformation.

evaluation. For novel view synthesis, we hold out every 8th frame as the test set, using the rest for training. For novel view and time synthesis, we follow [8] to exclude alternate frames from both training and testing for interpolation evaluation.

C. Comparison with State-of-the-Art Methods

We compare ST-GS against 11 existing methods: EndoNeRF [12], RoDynRF [6], Endo-4DGS [3], SurgicalGaussian [14], MApo [4], 4D-GS [16], EndoGS [18], EndoGaussian [5], StreamSplat [13], EndoPlanar [1], and DeGauss [11]. All methods are retrained on the same splits. For COLMAP-based methods, we run COLMAP with best-effort parameters, though it often fails on textureless tissue and specular highlights [12]. For COLMAP-free baselines, we use official implementations.

1) Novel View Synthesis: Table I reports quantitative results on eight surgical scenes. Our ST-GS consistently achieves the highest PSNR across all scenes, outperforming all NeRF-based and 3DGS-based competitors by substantial margins. SSIM scores further confirm superior perceptual quality. Rendering speed exceeds 100 FPS on RTX 4090, orders of magnitude faster than NeRF-based methods and competitive with the fastest 3DGS approaches. Training time is also remarkably low compared to methods requiring over 24 hours. Figures 3 and 4 show that existing methods suffer from blurring near tool boundaries and floating artifacts near cuts, whereas our method produces sharp edges, clean occlusion handling, and consistent geometry. This demonstrates MG-TPC’s effective decoupling

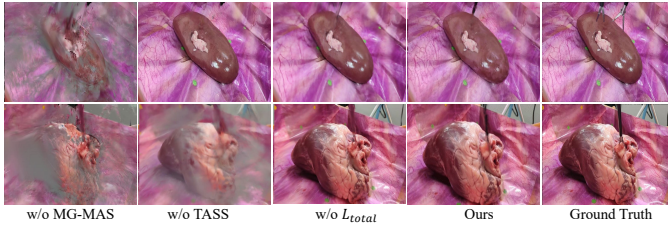


Fig. 7. Visual comparisons for ablation study on the EndoNeRF dataset. Removing MG-TPC, TASS, or the full loss function leads to severe artifacts, while our full model produces results closest to the ground truth.

of rigid tool motion from non-rigid tissue deformation, and TASS’s capability in handling topological changes.

TABLE III

ABLATION STUDIES. (A) ABLATION OF MG-TPC SUB-MODULES: PSNR, LPIPS, AND PER-GAUSSIAN DEFORMATION LATENCY g_{def} (NS). (B) ABLATION OF LOSS FUNCTIONS: PSNR, LPIPS, AND RMSE. THE FULL MODEL (OURS) YIELDS THE HIGHEST PSNR AND LOWEST LPIPS/RMSE, VALIDATING THE CONTRIBUTION OF EACH COMPONENT.

(a)MG-TPC			
	PSNR $\uparrow$	LPIPS $\downarrow$	g_{def} (ns)
Zernik Spectral Trajectory Field	23.51	0.125	149.41
Online Binary Tool Mask	25.48	0.090	98.89
Laplace-Beltrami Eigen-Relation	24.47	0.118	6.85
Scaling Hyperparameter	25.14	0.111	7.80
Cyclic Spatio-Temporal Evolution Paradigm	24.38	0.120	7.71
spectral-anchored depth consistency	26.19	0.060	8.78
Ours	29.41	0.057	5.63
(b)Loss function			
	PSNR $\uparrow$	LPIPS $\downarrow$	RMSE $\downarrow$
w/o $\mathcal{L}_{\text{photo}}$	17.49	0.853	28.763
w/o $\mathcal{L}_{\text{ssim}}$	25.33	0.067	11.658
w/o $\mathcal{L}_{\text{depth}}$	24.18	0.066	10.445
w/o $\sum_g \sum_{n,m} \ \mathbf{C}_{n,m}^{(g)}\ _1$	26.34	0.088	12.873
Ours	29.03	0.053	5.605

2) Novel View and Time Synthesis: Following [12], we evaluate temporal interpolation by sampling timestamps and excluding odd-indexed frames from training. Table II reports averaged results. Our ST-GS significantly outperforms both COLMAP-dependent methods and COLMAP-free baselines in PSNR and LPIPS. The temporal consistency metric tOF is also markedly lower (better), indicating smoother frame transitions. Figure 5 demonstrates that our method faithfully reconstructs continuous tissue motion and handles abrupt topological changes like incisions, while baselines produce blurring and geometric distortion at unseen timestamps. TASS is crucial here—it detects photometric error spikes and splits trajectories for accurate cutting boundary modeling.

3) Depth Map Analysis of MG-TPC Decoupling: To provide direct geometric evidence of decoupling, we visualize depth maps in Fig. 6. In scenes where a rigid instrument moves across tissue, methods without MG-TPC erroneously propagate tool motion into tissue depth, producing abrupt jumps and “floating” structures behind the instrument—indicating deformation field contamination. In contrast, our MG-TPC adaptively freezes gradients for occluded Gaussians via exponential attenuation (Eq. 4), interpolating depth from historical trajectory rather than erroneous photometric signals. As shown

TABLE IV

TASS ABLATION STUDY ON THE ENDONeRF CUTTING SCENES (AVERAGE OVER FOUR SEQUENCES). WE COMPARE OUR FULL MODEL WITH TASS AGAINST TWO ABLATED VARIANTS. g_{split} DENOTES THE EXTRA PER-GAUSSIAN COMPUTATIONAL OVERHEAD (IN NS) INCURRED BY THE SPLITTING STRATEGY. (FAS: FLOATING ARTIFACT SCORE)

Method	PSNR $\uparrow$	g_{split} (ns) $\downarrow$	LPIPS $\downarrow$	FAS $\downarrow$	RMSE(mm) $\downarrow$
w/o TASS	25.32	—	0.092	0.68	12.68
Naive TASS	26.51	12.4	0.072	0.31	8.93
Full TASS	27.21	8.6	0.053	0.12	4.68

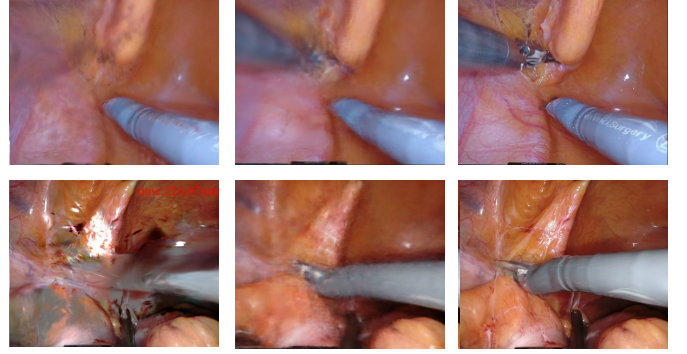


Fig. 8. Visual comparison of different TASS variants on the Cutting Prostate scene. *w/o TASS* produces severe blurring and geometric bleeding at the incision boundary. *Naive TASS* still exhibits visible residual artifacts. Our full TASS yields sharp and clean cut edges.

in Fig. 6, ST-GS depth maps maintain temporal continuity: the tissue retains a coherent depth profile before, during, and after occlusion, with no motion leakage from the tool. This confirms MG-TPC successfully decouples the two motion components at the geometric level, a prerequisite for high-fidelity reconstruction.

D. Ablation Study

We conduct comprehensive ablation studies on the EndoNeRF dataset to quantify the contribution of each component in our framework. Table ?? presents the quantitative results.

1) Effect of MG-TPC Sub-modules: Table ??(a) breaks down the influence of each element in MG-TPC. Replacing the Zernike spectral trajectory basis with conventional cubic Hermite splines (i.e., using the SplineGS backbone) significantly increases the per-Gaussian deformation latency g_{def} and degrades PSNR, confirming that our spectral representation offers superior efficiency and accuracy for surgical dynamics. Removing the online binary tool mask (i.e., disabling occlusion awareness) forces the network to treat tool motion as tissue deformation, resulting in severe geometric entanglement. Ablating the frequency-selective gradient mask (using uniform gradients regardless of occlusion) causes the control points behind tools to drift randomly, degrading both PSNR and LPIPS. The scaling hyperparameter γ controls the softness of gradient freezing: setting it to zero (no attenuation) or an excessively large value (hard freezing) both harm performance, with the optimal value yielding a balanced trade-off. Removing

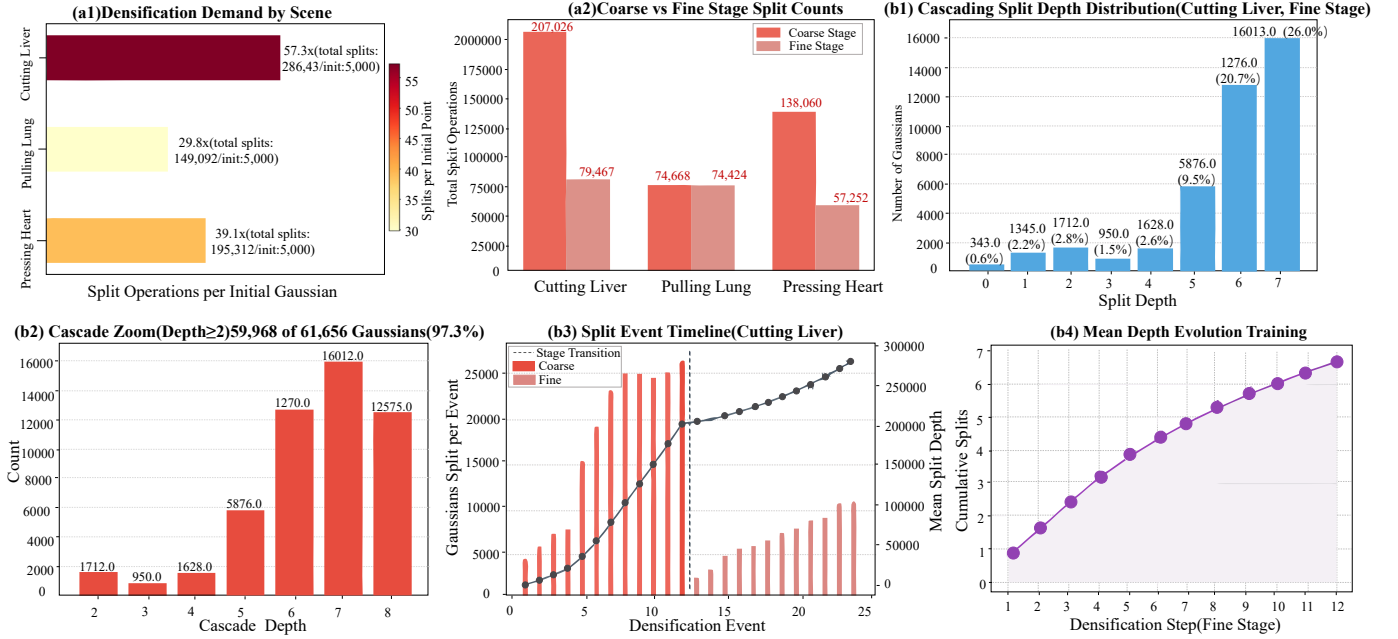


Fig. 9. Analysis of TASS efficacy. (a) Split ratio per initial Gaussian per scene (color-coded), with coarse/fine decomposition. (b) Split depth distribution: histogram, cascaded (depth ≥ 2), cumulative, and average depth evolution. TASS adaptively allocates deeper splits to complex cuts (Cutting Liver) and shallower ones to simpler motions (Pulling Lung, Pressing Heart).

the interpolation compensation mechanism (which relies on historical spline momentum during full occlusion) introduces temporal discontinuities. Finally, replacing accurate Gaussian projection with coarse bounding boxes introduces errors in occlusion ratio estimation. The full MG-TPC model with all components achieves the best overall performance.

2) Effect of Loss Functions: Table ??(b) shows the impact of each loss term. Without the masked photometric loss $\mathcal{L}_{\text{photo}}$ (using unmasked photo loss), the model catastrophically reconstructs tool regions as if they were tissue. Removing the SSIM loss $\mathcal{L}_{\text{ssim}}$ marginally reduces perceptual quality, confirming its role in preserving structural details in textureless areas. Ablating the depth-consistency loss $\mathcal{L}_{\text{depth}}$ produces floating artifacts and depth inaccuracies, underscoring that monocular depth priors are essential for resolving scale ambiguity. Removing the trajectory smoothness loss (or spectral sparsity in our full model) results in jittery motion, as spectral coefficients are free to produce high-acceleration artifacts. The full loss combination yields the highest PSNR and lowest LPIPS.

3) Effect of TASS and Depth Regularization: In additional experiments (not shown in tables due to space), we evaluate the TASS strategy separately. When TASS is disabled (allowing continuous splines to represent cutting boundaries), the model produces severe geometric bleeding and blurring at incision sites, with notable PSNR degradation across cutting scenes. The depth-consistency loss also exhibits a synergistic effect with TASS: without $\mathcal{L}_{\text{depth}}$, the newly spawned child primitives after a cut tend to drift apart in depth, creating unnatural gaps; incorporating $\mathcal{L}_{\text{depth}}$ stabilizes the bifurcation process. Visual ablation comparisons confirm that removing MG-TPC, TASS, or the full loss function leads to pronounced artifacts (floating matter, edge blurring, and structural distortions), while our complete model produces results closest to ground

truth. These ablations collectively validate the necessity of each proposed component.

E. Topology-Aware Spectral Splitting (TASS)

To assess the effectiveness of our proposed topology-aware spectral splitting (TASS) strategy for handling irreversible topological mutations in surgical scenes, we compare our full model against two ablated variants: (i) **w/o TASS**, where splitting is disabled and all Gaussians follow a single continuous Zernike spectral trajectory; (ii) **Naive TASS**, which employs a fixed photometric error threshold without three-frame continuity verification. Experiments are conducted on public, in-house, and hospital datasets covering diverse surgical scenarios with abrupt tissue dissections.

Table ?? reports the ablation results. Our full TASS achieves the best PSNR and LPIPS, outperforming both w/o TASS and Naive TASS. The floating artifact score (FAS) is substantially reduced, confirming effective suppression of geometric bleeding at incisions. Full TASS also lowers per-Gaussian splitting overhead compared to Naive TASS, indicating that adaptive thresholding avoids unnecessary splits. In terms of geometric accuracy, our model achieves the lowest RMSE, preserving millimeter-level fidelity under tissue rupture. These results demonstrate that TASS offers a favorable trade-off among quality, efficiency, and accuracy—critical for intraoperative reconstruction.

We further analyze splitting statistics across surgical scenes in Fig. ?. Fig. ??(a) presents the split ratio heatmap, where each bar denotes the average number of splits per initial Gaussian per sequence. Scenes with aggressive cutting exhibit significantly higher split ratios than pressing or pulling sequences, indicating that TASS activates more frequently during

abrupt structural ruptures. The right panel decomposes splits into coarse-stage and fine-stage. Notably, most splits occur in the coarse stage, where initial continuous Zernike spectral trajectories fail to fit discontinuous motion; fine-stage splits primarily refine boundaries.

Fig. ??(b) analyzes split depth distribution. The histogram in (b1) shows that most Gaussians undergo shallow splits (depth = 1), corresponding to a single bifurcation event. However, a non-negligible portion reaches depth ≥ 2 , indicating cascaded splitting in highly dynamic regions where tissue lips continue to separate. The cumulative distribution confirms that only a small fraction of Gaussians require depth ≥ 3 . Fig. ??(b4) plots the evolution of average split depth: it increases rapidly during early training and then plateaus, demonstrating that TASS quickly adapts to topological changes and stabilizes thereafter. Compared to the w/o TASS variant, our full TASS achieves substantial PSNR gains on cutting scenes and significantly reduces FAS. These analyses confirm that TASS reliably detects incisions and adaptively allocates splitting complexity according to the severity of topological mutations, enabling clean and accurate reconstruction of tissue boundaries.

V. CONCLUSION

To overcome occlusions, missing views, and tissue cutting in monocular endoscopic reconstruction, we propose ZT-GS, a Zernike spectral trajectory field 3DGS for dynamic reconstruction without pre-computed poses. ZSTF constructs continuous trajectories via spectral decomposition on a conformal disk for frequency disentanglement. Building upon this, MG-TPC decouples rigid instrument motion from tissue deformation under occlusion, while TASS detects tissue ruptures via spectral entropy and performs trajectory bifurcation. A depth-consistency loss anchors global scale and eliminates floaters, and a cyclic forward-backward paradigm generates the global dynamic model. Extensive experiments on EndoNeRF, in-house, and clinical datasets show that ZT-GS consistently outperforms state-of-the-art methods in novel-view synthesis, temporal consistency, and geometric accuracy, while achieving real-time rendering over 100 FPS. However, MG-TPC's reliance on accurate tool masks and TASS's false positives under motion blur or lighting changes. Future work will address these via self-supervised segmentation, adaptive thresholds, and online learning. Nevertheless, ZT-GS represents an important step toward COLMAP-free, physically-constrained 3D Gaussian splatting for complex surgical scenes.

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